



The Living World

Unit 1

Diversity in the Living World - Unit Introduction

Biology is the science of life forms and living processes. The living world comprises an amazing diversity of living organisms.

- Description of living organisms, including human beings, began much later in human history. Societies with an **anthropocentric view** of biology could register only limited progress in biological knowledge.
- Systematic and monumental description of life forms brought in, *out of necessity*, detailed systems of **identification, nomenclature and classification**.
- The biggest spin off of such studies was recognising that similarities are shared among living organisms **both horizontally and vertically**.
- That all present day living organisms are related to each other, and also to all organisms that ever lived on this earth, was a revelation that humbled man and led to **cultural movements for conservation of biodiversity**.

Profile

Ernst Mayr (1904 - 2004)

Item	Detail
Born	5 July 1904, in Kempten, Germany
Died	2004, at the age of 100
Position	Harvard University evolutionary biologist; joined Harvard's Faculty of Arts and Sciences in 1953, retired in 1975, assuming the title Alexander Agassiz Professor of Zoology Emeritus
Called	'The Darwin of the 20th century' ; one of the 100 greatest scientists of all time
Career span	Nearly 80-year career ; research spanned ornithology, taxonomy, zoogeography, evolution, systematics, and the history and philosophy of biology
Credited with	Almost single-handedly making the origin of species diversity the central question of evolutionary biology that it is today; pioneering the currently accepted definition of a biological species
Triple crown of biology	Balzan Prize (1983), International Prize for Biology (1994), Crafoord Prize (1999)

1.0

What Indeed is Life? - The Question Behind the Chapter

The wide range of living types is amazing. Organisms occupy extraordinary habitats - **cold mountains, deciduous forests, oceans, fresh water lakes, deserts or hot springs**. The beauty of a galloping horse, of the migrating birds, the valley of flowers or the attacking shark evokes awe and a deep sense of wonder. So does ecological **conflict and cooperation** among members of a population and among populations of a community, and even the **molecular traffic inside a cell**.

The very range of organisms - in terms of **size, colour, habitat, physiological and morphological features** - makes us seek the defining characteristics of living organisms. The question 'what indeed is life?' carries **two implicit questions**:

1

The technical question - what is living, as opposed to the non-living? Science answers this one.

2

The philosophical question - what is the purpose of life? *As scientists, we shall not attempt answering the second question.*

1.1 Diversity in the Living World

Several organisms **cannot be seen with the naked eye**, yet they are all around you. Increase the area you observe and the range and variety of organisms increases; visit a **dense forest** and you would *probably* see a much greater number and kinds of living organisms.

- **Species** - each different kind of plant, animal or organism that you see represents a species.
- **Biodiversity** - the number and types of organisms present on earth. The number of species that are known and described range between **1.7-1.8 million**.
 - Millions of plants and animals have been identified and described, but a **large number still remains unknown** - as we explore new areas, and even old ones, new organisms are **continuously being identified**.
 - Taxonomic studies of various species of plants and animals are useful in **agriculture, forestry, industry**, and in general for knowing our **bio-resources** and their diversity.

1.1 Nomenclature and Identification - Why Local Names Fail

We know the plants and animals in our own area by their **local names**, and these vary from place to place, **even within a country** - which is exactly the confusion that makes talking about an organism impossible. Hence naming must be standardised so that a particular organism is known by the **same name all over the world**.

- **Nomenclature** - the process of standardising the naming of living organisms. It is *only* possible when the organism is described correctly and we know to what organism the name is attached.
- **Identification** - describing the organism correctly and knowing which organism a name belongs to. Identification must therefore come before naming.

A number of scientists have established procedures to assign a scientific name to each known organism, acceptable to biologists all over the world:

Code	Full form	Applies to
ICBN	International Code for Botanical Nomenclature - scientific names based on agreed principles and criteria	Plants
ICZN	International Code of Zoological Nomenclature - evolved by animal taxonomists	Animals

- Scientific names ensure that each organism has **only one name**.
- A description of any organism should enable people **in any part of the world** to arrive at the same name.
- They also ensure that such a name **has not been used for any other known organism**.

1.1 Binomial Nomenclature

Biologists follow universally accepted principles to provide scientific names. Each name has **two components** - the **Generic name** and the **specific epithet**. This system of providing a name with two components is called **Binomial nomenclature**, given by **Carolus Linnaeus** and practised by biologists all over the world; the two word format was found convenient.

Worked example - the scientific name of mango is *Mangifera indica*: *Mangifera* represents the **genus**, while *indica* is a particular species, or a **specific epithet**.

Rule	What it says	Illustration
1. Language	Biological names are <i>generally</i> in Latin and written in italics . They are Latinised or derived from Latin <i>irrespective of their origin</i> .	<i>Mangifera indica</i>
2. Order of words	The first word represents the genus ; the second component denotes the specific epithet .	<i>Mangifera</i> = genus, <i>indica</i> = specific epithet

Rule	What it says	Illustration
3. Handwriting vs print	Both words, when handwritten , are separately underlined ; when printed they are in italics - to indicate their Latin origin.	<u>Mangifera</u> <u>indica</u>
4. Capitalisation	The first word denoting the genus starts with a capital letter ; the specific epithet starts with a small letter .	<i>Mangifera indica</i> (not <i>Mangifera Indica</i>)
5. Author citation	The name of the author appears after the specific epithet , i.e. at the end of the biological name, in an abbreviated form . It indicates who first described the species.	<i>Mangifera indica</i> Linn. - first described by Linnaeus

1.1

Classification, Taxonomy and Systematics

Since it is **nearly impossible** to study all the living organisms, some means is necessary to make the study possible - this process is **classification**.

- **Classification** - the process by which anything is grouped into **convenient categories** based on some **easily observable characters**.

A group name (dogs, mammals, wheat) instantly recalls a set of shared characters; this is the basis of classification.

- **Taxa** - the scientific term for such convenient categories. Taxa can indicate categories at **very different levels**: 'Plants', 'Wheat', 'animals', 'mammals' and 'dogs' are all taxa - yet a dog is a mammal and mammals are animals, so they represent taxa at different levels.

Based on characteristics, all living organisms can be classified into different taxa - and **this process of classification is taxonomy**. The basis of **modern taxonomic studies** is: **external and internal structure**, along with the **structure of cell, development process** and **ecological information** of organisms.

Four processes are **basic to taxonomy**:

- 1 **Characterisation** - record the characters of the organism.
- 2 **Identification** - establish which organism it is.
- 3 **Classification** - place it in convenient categories (taxa).
- 4 **Nomenclature** - give it its universally valid two-part name.

Taxonomy is **not something new**. Human beings have always been interested in knowing more about the various kinds of organisms, particularly with reference to their own use: in early days they needed sources for their basic needs of **food, clothing and shelter**, so the **earliest classifications were based on the 'uses'** of various organisms.

- **Systematics** - the branch of study of the **relationships** among different kinds of organisms, not merely of their kinds and diversities. The word is derived from the Latin word '**systema**', meaning systematic arrangement of organisms; **Linnaeus used *Systema Naturae*** as the title of his publication.
- The scope of systematics was **later enlarged** to include **identification, nomenclature and classification**.
- Systematics takes into account **evolutionary relationships** between organisms.

① **[NOTE]** Do not blur the four terms. **Identification** = which organism is this? **Nomenclature** = what is its one valid name? **Classification** = grouping into taxa on easily observable characters. **Taxonomy** = the whole process of classifying organisms into taxa. **Systematics** is the widest: identification + nomenclature + classification, plus evolutionary relationships.

Classification is **not a single step process** but involves a **hierarchy of steps**, in which each step represents a **rank** or **category**. Since the category is a part of the overall taxonomic arrangement, it is called the **taxonomic category**, and all categories together constitute the **taxonomic hierarchy**.

● **Taxon** (pl.: **taxa**) - each category, referred to as a **unit of classification**, which in fact represents a **rank**.

Example: **insects** represent a group of organisms sharing common features like **three pairs of jointed legs**. That makes insects **recognisable concrete objects** which can be classified, and thus they were given a rank or category. Groups represent category; category further denotes rank; each rank or taxon represents a unit of classification.

① **[NOTE]** These taxonomic groups/categories are **distinct biological entities and not merely morphological aggregates** - a favourite one-line NEET statement.

Taxonomical studies of all known organisms have led to the development of **common categories: kingdom, phylum or division (for plants), class, order, family, genus and species**. All organisms, including those in the plant and animal kingdoms, have **species as the lowest category**. To place an organism in these categories, the basic requirement is **knowledge of the characters** of an individual or group of organisms - this helps in identifying **similarities and dissimilarities** among individuals of the same kind of organisms as well as of other kinds.

1.2.1

Species

Taxonomic studies consider a group of individual organisms with **fundamental similarities** as a **species**. One should be able to distinguish one species from the other **closely related species** based on the **distinct morphological differences**. Ernst Mayr pioneered the currently accepted definition of a biological species.

- In *Mangifera indica*, *Solanum tuberosum* (potato) and *Panthera leo* (lion), the names *indica*, *tuberosum* and *leo* represent the **specific epithets**, while the first words *Mangifera*, *Solanum* and *Panthera* are **genera** and represent another **higher level of taxon** or category.
- Each genus *may* have **one or more than one** specific epithets representing different organisms, but having **morphological similarities**: *Panthera* has another specific epithet called *tigris*, and *Solanum* includes species like *nigrum* and *melongena*.
- Human beings belong to the species *sapiens*, grouped in the genus *Homo* - so the scientific name for human being is *Homo sapiens*.

1.2.2

Genus

Genus comprises a group of **related species** which has **more characters in common** in comparison to species of other genera; genera are **aggregates of closely related species**.

- Potato and brinjal are **two different species** but both belong to the genus *Solanum*.
- Lion (*Panthera leo*), leopard (*P. pardus*) and tiger (*P. tigris*), with several common features, are all species of the genus *Panthera*. This genus **differs from** another genus *Felis*, which includes cats.

1.2.3

Family

Family, the next category, has a group of **related genera** with **still less number of similarities** as compared to genus and species. Families are characterised on the basis of **both vegetative and reproductive features** of plant species.

Kingdom side	Genera placed together	Family
Plants	<i>Solanum</i> , <i>Petunia</i> , <i>Datura</i> (three different genera)	Solanaceae
Animals	<i>Panthera</i> (lion, tiger, leopard) put along with <i>Felis</i> (cats)	Felidae
Animals	The dog line - observe a cat and a dog and you find some similarities and some differences as well, so they are separated into two different families	Canidae (dog) vs Felidae (cat)

Categories like **species, genus and families** are based on a **number of similar characters**. *Generally, order* and other higher taxonomic categories are identified based on the **aggregates of characters**.

- Order, being a higher category, is the **assemblage of families** which exhibit **a few similar characters**; these similar characters are **less in number** as compared to different genera included in a family.
- Plants: families like **Convolvulaceae** and **Solanaceae** are included in the order **Polymoniales**, *mainly* based on the **floral characters**.
- Animals: the order **Carnivora** includes families like **Felidae** and **Canidae**.

This category includes **related orders**. For example, order **Primata** comprising **monkey, gorilla and gibbon** is placed in class **Mammalia**, along with order **Carnivora** that includes animals like **tiger, cat and dog**. **Class Mammalia has other orders also**.

Classes comprising animals like **fishes, amphibians, reptiles, birds** along with **mammals** constitute the next higher category called **Phylum**. All these, based on the common features like presence of **notochord** and **dorsal hollow neural system**, are included in phylum **Chordata**. In case of plants, classes with **a few similar characters** are assigned to a higher category called **Division**.

All animals belonging to various **phyla** are assigned to the **highest category** called **Kingdom Animalia** in the classification system of animals. The **Kingdom Plantae**, on the other hand, is **distinct**, and comprises all plants from various **divisions**. Henceforth these two groups are referred to as **animal and plant kingdoms**.

The taxonomic categories from **species to kingdom** in **ascending order**, starting with species, are: **Species, Genus, Family, Order, Class, Phylum or Division, Kingdom**. These are **broad categories**; taxonomists have also developed **sub-categories** in this hierarchy to facilitate more sound and scientific placement of various taxa.

Rank (ascending)	What it groups	Chapter example
Kingdom	The highest category ; all phyla of animals, all divisions of plants	Kingdom Animalia; Kingdom Plantae
Phylum or Division	Phylum - classes of animals with common features; Division - plant classes with a few similar characters	Chordata (notochord, dorsal hollow neural system); Angiospermae
Class	Related orders	Mammalia (has other orders also); Insecta
Order	Assemblage of families exhibiting a few similar characters	Carnivora; Polymoniales; Primata
Family	Related genera , with still less number of similarities	Felidae; Canidae; Solanaceae
Genus	Aggregate of closely related species	<i>Panthera</i> ; <i>Solanum</i>
Species	The lowest category ; individuals with fundamental similarities	<i>Panthera leo</i> ; <i>Homo sapiens</i>

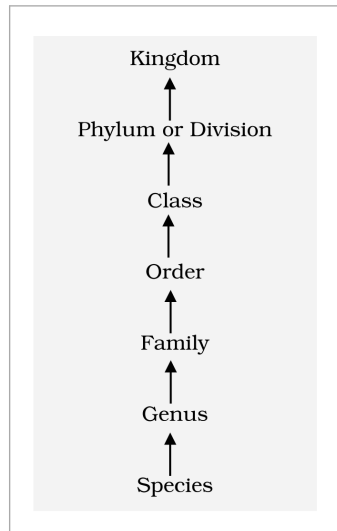


Fig. 1.1 - Taxonomic categories showing hierarchial arrangement in ascending order.

Reading the hierarchy tells you the **basis of arrangement**:

- As we go **higher** from species to kingdom, the **number of common characteristics goes on decreasing**.
- **Lower the taxa, more are the characteristics** that the members within the taxon share.
- **Higher the category, greater is the difficulty** of determining the relationship to other taxa **at the same level** - hence the problem of classification becomes **more complex**.

☆ **[MEMORY AID - not in NCERT]** Ascending order **Species to Kingdom**: *Some Girls Find Our Class Pretty Kind - Species, Genus, Family, Order, Class, Phylum, Kingdom*. Read it backwards for descending order.

Table 1.1 Organisms with their Taxonomic Categories

Table 1.1 indicates the taxonomic categories to which some common organisms like **housefly, man, mango and wheat** belong.

Common Name	Biological Name	Genus	Family	Order	Class	Phylum/Division
Man	<i>Homo sapiens</i>	<i>Homo</i>	Hominidae	Primata	Mammalia	Chordata
Housefly	<i>Musca domestica</i>	<i>Musca</i>	Muscidae	Diptera	Insecta	Arthropoda
Mango	<i>Mangifera indica</i>	<i>Mangifera</i>	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae
Wheat	<i>Triticum aestivum</i>	<i>Triticum</i>	Poaceae	Poales	Monocotyledonae	Angiospermae

① **[NOTE]** Read Table 1.1 column-wise for the two plant entries: mango is **Dicotyledonae** and wheat is **Monocotyledonae**, yet both are **Angiospermae** - the class differs while the division is the same.

Recap Quick Recap

- The living world is **rich in variety**. Millions of plants and animals have been identified and described, but a **large number still remains unknown**. The range of organisms in **size, colour, habitat, physiological and morphological features** is what makes us seek the defining characteristics of living organisms. Known and described species: **1.7-1.8 million**.
- To facilitate the study of kinds and diversity of organisms, biologists evolved **rules and principles for identification, nomenclature and classification**; the branch of knowledge dealing with these aspects is **taxonomy**.

- The basics of taxonomy - identification, naming and classification - are **universally evolved under international codes: ICBN** for plants, **ICZN** for animals.
- Based on **resemblances and distinct differences**, each organism is identified and assigned a correct scientific/biological name comprising **two words** as per the **binomial system of nomenclature** (Carolus Linnaeus): capitalised genus, small-letter specific epithet, Latin, italics in print and separately underlined when handwritten, with the abbreviated author name at the end.
- An organism **occupies a place or position** in the system of classification. There are many **categories/ranks**, generally referred to as **taxonomic categories** or **taxa**, and all the categories constitute a **taxonomic hierarchy** - species (lowest) to kingdom (highest).
- Taxonomic studies of species of plants and animals are useful in **agriculture, forestry, industry** and for knowing our **bio-resources** and their diversity. **Systematics** adds **evolutionary relationships** to identification, nomenclature and classification.

Appendix

Terms Used in the Exercises

- **Why classification systems keep changing (Q2).** Three chapter facts explain it: (i) as we explore new areas, and even old ones, **new organisms are continuously being identified**, so the arrangement must accommodate them; (ii) taxonomists keep developing **sub-categories** in the hierarchy to facilitate more sound and scientific placement of various taxa; (iii) **systematics takes evolutionary relationships into account**, and the basis of modern taxonomic studies includes internal structure, cell structure, development process and ecological information - as this evidence improves, the grouping is revised.
- **What identification of individuals and populations gives us (Q4).** Identification is what makes **nomenclature possible at all** - a name can be attached only to a correctly described organism. It yields a single universally valid name per organism, so that a description lets people in **any part of the world** arrive at the same name and no name is reused for another known organism; and it is how the count of **known and described species (1.7-1.8 million)**, i.e. our record of **biodiversity**, is built.
- **Answer keys the chapter fixes directly.** Q5 - the correctly written name is *Mangifera indica* (genus capitalised, specific epithet in small letters). Q7 - the correct sequence is **Species, Genus, Order, Phylum**, i.e. option (c), because the ascending order is Species, Genus, Family, Order, Class, Phylum or Division, Kingdom. Q6 - a **taxon** is a unit of classification representing a rank, e.g. *Homo* (genus), Hominidae (family), Primata (order), Mammalia (class), Chordata (phylum).